



Boron/zinc exchange for the conjugate arylation of unsaturated Meldrum's acid derivatives

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ABSTRACT

An approach for the conjugate arylation reaction of unsaturated Meldrum's acid derivatives is reported. The reaction takes place through a Boron-Zinc exchange reaction between arylboronic acids and diethylzinc and the subsequent transfer of thus generated reactive aryl groups to the β -position of the double bond of the Meldrum's derivatives. The compounds obtained are valuable synthetic intermediates and can give rise to bioactive molecules such as β,β -disubstituted carboxylic acids and amides.

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1. Introduction

Meldrum's acid and its unsaturated derivatives are important building blocks, widely employed in organic synthesis [1]. Several reports can be found in the literature in which Meldrum's acid alkyl(aryl)idenes have served as substrates for a wide range of reactions with subsequent applications in the total synthesis of natural products and biologically relevant molecules [2–8]. For example, the synthesis of taiwanaquinol B [9], ar-turmerone and α -curcumene [10], and coumarin derivatives [11,12] have all been accomplished through the use of Meldrum's acid alkylidenes as a key intermediate in the synthesis endeavor. The interest in using unsaturated Meldrum's acid derivatives arises due to their excellent ability to act as electrophiles for conjugate addition reactions with a range of nucleophiles. Conjugate additions of organometallic species to Meldrum's alkyl(aryl)idenes stand out as a valuable tool to forge carbon-carbon bonds in these systems. Among them, the

most studied reaction is the conjugate addition of alkynes, that can be catalyzed by copper [13], zinc [14], and rhodium [15], or the direct alkynylation using zinc- [16], aluminum-, and Grignard-alkynilides [17]. Copper-catalyzed conjugate alkylations using dialkylzinc reagents [18] as well as the Rh-catalyzed addition of vinylstannanes [19] have also been reported. On the other hand, the conjugate addition of aryl nucleophiles to Meldrum's derivatives have been less studied. Reported examples include the Rh-catalyzed addition of dioxaborinanes [20], and copper-catalyzed Grignard addition [21]. These methods, however, present some limitations; the former demands the use of an alkylidene with of an aryl group bearing a coordinating oxygen atom at the *ortho* position, while the latter is limited to only a few variations of substituents at both reactants (Scheme 1A).

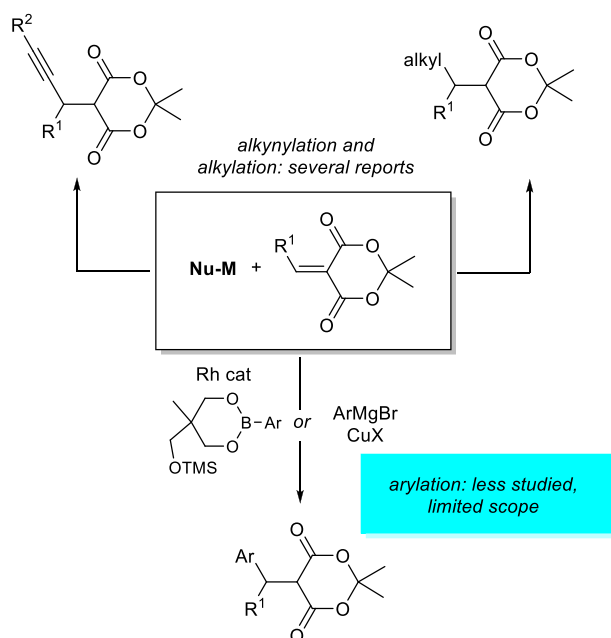
As part of the research interests of our group, over the past few years we have studied the boron-zinc exchange reaction between arylboronic acids and diethylzinc for the generation of transferable aryl groups. The thus generated mixed arylzinc reagents have been used for the mild and selective addition to a variety of carbonyl electrophiles [22,23]. With this in mind, we wondered if this strategy might be examined towards a Michael-type acceptor. Herein we report our results of a mild, transition-metal-free conjugate

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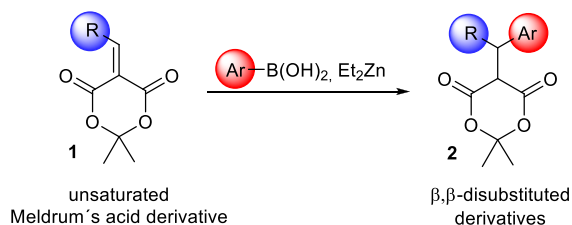
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A. Conjugate addition of organometallic nucleophiles to Meldrum's alkylidenes



B. This work: conjugate arylation enabled by a B/Zn exchange



Scheme 1. Precedents and this work.

arylation of unsaturated Meldrum's acid derivatives enabled by the a B/Zn exchange reaction (Scheme 1B). The ensuing products are versatile compounds that can be readily converted to β,β -disubstituted carboxylic acids and amides, which are structural units present in a number of biologically relevant molecules.

2. Results and discussion

At the outset, we selected the alkylidene **1a** and phenylboronic acid as model substrates for the optimization of the reaction conditions. The reactive arylating agent was generated *in situ* by the reaction of diethylzinc and phenylboronic acid. Initial screenings have been focused on the optimization of the reaction solvent and stoichiometry (Table 1). In our first attempt, using 1.2 equiv. of phenylboronic acid and 3.6 equiv. of diethylzinc in toluene as the solvent, the desired product was not formed (entry 1), mainly due to the low solubility of the Meldrum's arylidene in the reaction mixture. When THF was used to solubilize the arylidene **1a**, the desired arylated product **2a** was isolated in 78% yield (entry 2). Increasing of the amount of Et₂Zn led to a decreased yield of product **2a** (entry 3). Since we have observed that the arylidene was not completely consumed in the initial experiments, we decided to increase the amount of the arylating agent. Thus, when the reaction was performed in the presence of 1.6 equiv. of the mixed arylzinc reagent, the desired product was obtained in an increased 86% yield

Table 1
Optimization of the reaction conditions^a.

Entry	PhB(OH) ₂ (equiv.)	Et ₂ Zn (equiv.)	Time (h)	Solvent	Yield (%) ^b
1	1.2	3.6	4	toluene	NR
2	1.2	3.6	2	toluene/THF	78
3	1.2	4.0	2	toluene/THF	60
4	1.6	4.8	2	toluene/THF	86
5	2.4	7.2	2	toluene/THF	82
6	1.2	3.6	4	toluene/DCM	58
7	1.2	3.6	2	toluene/DMF	14
8	1.2	3.6	2	toluene/CPME	49
9	1.2	3.6	2	toluene/MTBE	46
10	1.2	3.6	4	toluene/THF	86
11	1.6	4.8	4	toluene/THF	94
12	2.4	7.2	4	toluene/THF	95

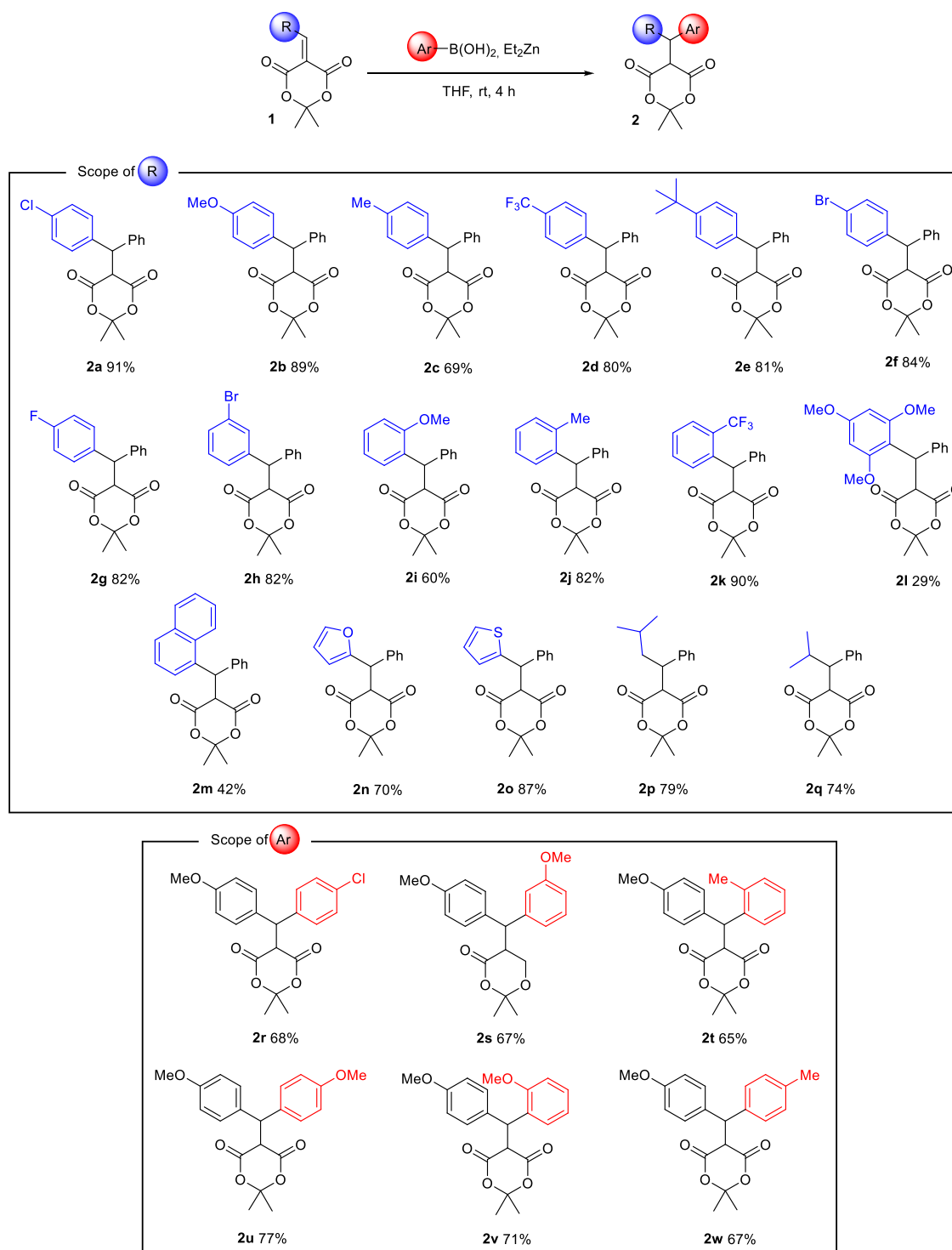
^a The reaction was performed on the 0.3 mmol scale. Diethylzinc was used as a 1.5 M solution in toluene.

^b Yield determined by NMR spectroscopy using mesitylene as the internal standard.

(entry 4). Increasing the amount of the arylating agent even further did not provide better yields (entry 5). Additional solvent screening was carried out and revealed that THF outperforms all other solvents investigated (entries 6–9). To further optimize the yield of the reaction, we studied the reaction time combined with different amounts of nucleophile (entries 10–12). In this study, we have found that the best yields have been achieved when 1.6 equiv. of the mixed arylzinc reagent (generated from 1.6 equiv. phenylboronic acid and 4.8 equiv. of Et₂Zn) was used (entry 11).

With suitable reaction conditions in hands, we investigated the scope of the reaction using a broader range of Michael acceptors and phenylboronic acid as the aryl source (Scheme 2). The reaction proceeded efficiently for a number of substrates bearing different groups with electron-donating and electron-withdrawing effects, at *ortho*, *meta* or *para* positions. For example, the arylation of substrates bearing *p*-Cl, *p*-OMe, *p*-CF₃, *o*-Me and *m*-Br were successfully used and the corresponding β,β' -arylated derivatives **2a**, **2b**, **2d**, **2j**, and **2h** were obtained in very good yields (80–91%). Steric effects have also been examined and the reaction is equally efficient when *ortho* substituents are present at the starting material. Hence, the transfer of a phenyl group to the alkylidenes with *ortho*-methoxy and *ortho*-trifluoromethyl groups provided the desired products **2i** and **2k** in 60% yield and 90% yield, respectively. The bulkier 1-naphthyl substituent at the β -position is also well tolerated and the product **2m** was obtained in 42% yield. Conversely, the presence of two *ortho*-methoxy groups causes a more severe steric hindrance and a decrease in the yield of product **2l**. Nevertheless, despite the more congested environment, the arylated product **2l** was still formed in 29% yield. The arylation of arylidenes substituted with heterocyclic side chains 2-furyl and 2-thienyl was also performed and the compounds **2n** and **2o** were obtained in 70% and 87% yield, respectively. Finally, the arylation of alkylidenes was also possible and product **2p** and **2q** bearing aliphatic side chains was also achieved in good yields.

We next turned our attention to the scope of the boronic acid reaction partner (Scheme 2). The conjugate arylation reaction was performed with the arylidene **1b** and both boronic acids bearing



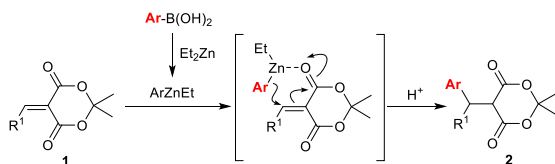
Scheme 2. Scope of the aryl transfer reaction to unsaturated Meldrum's acid derivatives.

electron-withdrawing and electron-donating groups were successfully used (**2r-w**). The presence of substituents with different electronic effects at the aryl group, such as *p*-Cl (**2r**), *m*-OMe (**2s**), *o*-Me (**2t**), *p*-OMe (**2u**), *o*-OMe (**2v**), and *p*-Me (**2w**), was well tolerated, and the products have been obtained in good yields.

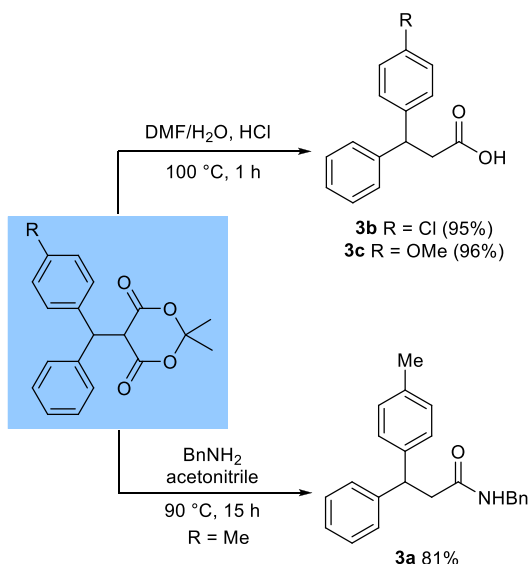
The mechanism of the reaction is presumed to occur through an initial coordination of the electron deficient zinc atom from the ArZnEt (which was generated *in situ* by the reaction of the arylboronic acid with diethylzinc [24]) to the carbonyl group, followed

by an intramolecular delivery of the aryl group [25] by a conjugate addition^{22c,d,f} (Scheme 3).

To highlight the utility of the reaction developed, derivatization reactions of the β,β -diarylated products have been performed (Scheme 4). Initially, we have carried out the hydrolysis reaction of the compounds **2a** and **2b**, and the β -disubstituted carboxylic acids **3b-c** were obtained in excellent yields. Additionally, we have reacted **2c** with benzylamine and the corresponding amide **3a** was isolated in 81% yield.



Scheme 3. Mechanistic hypothesis.



Scheme 4. Derivatization of the arylated products.

3. Conclusion

In summary, we have developed an efficient approach for the conjugate arylation reaction of unsaturated Meldrum's acid derivatives. The reaction showed good functional group tolerance and the arylated products were obtained in good yields, without the need of using additional transition metal catalysts. The compounds obtained are valuable synthetic intermediates and can give rise to bioactive molecules such as β -disubstituted carboxylic acids and amides. This methodology is operationally simple and should be useful for the synthesis of a variety of compounds thus opening a perspective for the use of the mixed zinc species in conjugate addition reactions. Importantly, the use of mildly nucleophilic organozinc reagents opens up the possibility of the development of an enantioselective version of the reaction which is otherwise more difficult to achieve by using more nucleophilic Grignard reagents.

4. Experimental section

4.1. General information

NMR spectra were recorded in CDCl_3 solution either on a 400 or 500 MHz spectrometer at room temperature. Chemical shifts (δ) are given in parts per million from the peak of tetramethylsilane ($\delta = 0.00$ ppm) as internal standard in ^1H NMR or from the solvent peak of CDCl_3 ($\delta = 77.0$ ppm) in ^{13}C NMR. ESI-QTOF-MS measurements were performed in the positive ion mode (m/z 50–2000 range). IR spectra were obtained on a FTIR-ATR instrument. Flash chromatography was performed using silica gel (230–400 mesh). Thin layer chromatography (TLC) was performed using supported silica gel GF₂₅₄, 0.25 mm thickness. For visualization, TLC plates were either placed under UV light 254 nm, stained with iodine

vapor or treated with vanillin solution followed by heating. Air- and moisture-sensitive reactions were conducted in dry glassware system equipped with tightly fitted rubber septa and under a positive pressure of dry argon. Meldrum's acid and unsaturated derivatives were synthesized according to literature procedures [26,27].

4.2. General procedure for the aryl transfer reaction

A 1.5 M solution of Et_2Zn in toluene (4.8 equiv., 1.44 mmol, 0.96 mL) was slowly added to a solution of arylboronic acid (1.6 equiv., 0.48 mmol) in dry toluene (1 mL) under argon. The mixture was stirred at 60 °C for 1 h and then cooled to 0 °C. Subsequently, a solution of the unsaturated Meldrum's acid derivative (0.3 mmol) in dry THF (2 mL) was added at 0 °C and the reaction mixture kept stirring at room temperature for 4 h. After this, saturated aqueous of NH_4Cl (5 mL) was carefully added at 0 °C. The product was extracted with ethyl acetate (3×10 mL). The combined organic layers were dried with MgSO_4 , filtered, the mixture was concentrated under reduced pressure and the crude was purified by flash chromatography on silica gel with hexane/ethyl acetate.

4.3. 5-((4-chlorophenyl)(phenyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2a**) [28]

White solid. **Yield:** 94.1 mg, 91%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). ^1H NMR (400 MHz, CDCl_3) δ 7.34–7.22 (m, 9H), 5.37 (d, $J = 2.7$ Hz, 1H), 4.27 (d, $J = 2.7$ Hz, 1H), 1.76 (s, 3H), 1.57 (s, 3H). ^{13}C NMR (100 MHz, CDCl_3) δ 164.5, 164.4, 139.6, 138.6, 133.0, 130.7, 129.1, 128.6, 128.5, 127.3, 105.2, 51.1, 48.3, 28.2, 27.5.

4.4. 5-((4-methoxyphenyl)(phenyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2b**) [28]

White solid. **Yield:** 90.8 mg, 89%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 75:25). ^1H NMR (400 MHz, CDCl_3) δ 7.32–7.28 (m, 4H), 7.27–7.21 (m, 3H), 6.86–6.80 (m, 2H), 5.34 (d, $J = 2.7$ Hz, 1H), 4.28 (d, $J = 2.7$ Hz, 1H), 3.78 (s, 3H), 1.74 (s, 3H), 1.52 (s, 3H). ^{13}C NMR (100 MHz, CDCl_3) δ 164.9, 164.7, 158.6, 140.4, 132.0, 130.5, 128.9, 128.4, 127.0, 113.7, 105.1, 55.2, 51.2, 48.5, 28.2, 27.6.

4.5. 2,2-Dimethyl-5-(phenyl(*p*-tolyl)methyl)-1,3-dioxane-4,6-dione (**2c**)

White solid (mp: 120 °C). **Yield:** 67.07 mg, 69%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). ^1H NMR (400 MHz, CDCl_3) δ 7.32–7.28 (m, 4H), 7.27–7.22 (m, 1H), 7.21–7.16 (m, 2H), 7.14–7.08 (m, 2H), 5.35 (d, $J = 2.8$ Hz, 1H), 4.29 (d, $J = 2.8$ Hz, 1H), 2.32 (s, 3H), 1.73 (s, 3H), 1.51 (s, 3H). ^{13}C NMR (100 MHz, CDCl_3) δ 164.8, 164.7, 140.2, 136.9, 136.7, 129.1, 129.08, 129.07, 128.4, 127.0, 105.1, 51.2, 48.8, 28.2, 27.6, 21.0. IR (ν_{max} , cm^{-1}): 2918, 1789, 1743, 1386, 1362, 1304, 1008, 723, 700. HRMS (ESI⁺): exact mass calculated for $[\text{M}+\text{H}]^+$ ($\text{C}_{20}\text{H}_{21}\text{O}_4$) requires m/z 325.1440, found: m/z 325.1439.

4.6. 2,2-Dimethyl-5-(phenyl(4-(trifluoromethyl)phenyl)methyl)-1,3-dioxane-4,6-dione (**2d**) [28]

White solid. **Yield:** 90.7 mg, 80%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). ^1H NMR (400 MHz, CDCl_3) δ 7.58–7.53 (m, 2H), 7.46–7.40 (m, 2H), 7.37–7.27 (m, 5H), 5.45 (d, $J = 2.7$ Hz, 1H),

4.31 (d, $J = 2.7$ Hz, 1H), 1.78 (s, 3H), 1.59 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 164.4, 164.3, 144.2 (q, $J = 1.2$ Hz), 139.2, 129.6, 129.3 (q, $J = 32.6$ Hz), 129.2, 128.7, 127.5, 125.3 (q, $J = 3.8$ Hz), 124.1 (q, $J = 272.0$ Hz), 105.3, 51.0, 48.5, 28.2, 27.5.

4.7. 5-((4-(*tert*-butyl)phenyl)(phenyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2e**) [29]

White solid. **Yield:** 88.9 mg, 81%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). **¹H NMR** (400 MHz, CDCl₃) δ 7.34–7.20 (m, 9H), 5.35 (d, $J = 2.7$ Hz, 1H), 4.30 (d, $J = 2.7$ Hz, 1H), 1.73 (s, 3H), 1.48 (s, 3H), 1.30 (s, 9H). **¹³C NMR** (100 MHz, CDCl₃) δ 164.8, 164.78, 149.9, 140.2, 136.9, 129.2, 128.8, 128.3, 127.0, 125.3, 105.1, 51.2, 48.8, 34.4, 31.3, 28.2, 27.7.

4.8. 5-((4-bromophenyl)(phenyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2f**) [28]

White solid. **Yield:** 102.4 mg, 84%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). **¹H NMR** (400 MHz, CDCl₃) δ 7.44–7.40 (m, 2H), 7.35–7.24 (m, 5H), 7.21–7.17 (m, 2H), 5.35 (d, $J = 2.7$ Hz, 1H), 4.28 (d, $J = 2.7$ Hz, 1H), 1.76 (s, 3H), 1.57 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 164.5, 164.4, 139.5, 139.1, 131.4, 131.0, 129.1, 128.6, 127.3, 121.2, 105.2, 50.9, 48.3, 28.2, 27.5.

4.9. 5-((4-fluorophenyl)(phenyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2g**) [28]

White solid. **Yield:** 80.7 mg, 82%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). **¹H NMR** (400 MHz, CDCl₃) δ 7.35–7.24 (m, 7H), 7.02–6.95 (m, 2H), 5.38 (d, $J = 2.7$ Hz, 1H), 4.28 (d, $J = 2.7$ Hz, 1H), 1.76 (s, 3H), 1.55 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 164.7, 164.5, 163.8 (d, $J = 246.2$ Hz), 139.9, 135.7 (d, $J = 3.2$ Hz), 131.0 (d, $J = 7.9$ Hz), 129.0, 128.5, 127.2, 115.2, (d, $J = 21.3$ Hz), 105.2, 51.2, 48.3, 28.2, 27.6.

4.10. 5-((3-bromophenyl)(phenyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2h**)

White Solid (mp: 119 °C). **Yield:** 95.2 mg, 82%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). **¹H NMR** (400 MHz, CDCl₃, δ ppm): δ 7.47 (t, $J = 1.8$ Hz, 1H), 7.40–7.36 (m, 1H), 7.35–7.22 (m, 6H), 7.17 (t, $J = 7.8$ Hz, 1H), 5.35 (d, $J = 2.6$ Hz, 1H), 4.28 (d, $J = 2.6$ Hz, 1H), 1.76 (s, 3H), 1.57 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 164.4, 164.3, 142.5, 139.2, 132.2, 130.2, 129.8, 129.1, 128.6, 127.9, 127.4, 122.4, 105.2, 51.0, 48.4, 28.2, 27.5. **IR** ($\nu_{\max}$, cm⁻¹): 1785, 1748, 1309, 1292, 1187, 1053, 1013, 920, 786, 720, 697. **HRMS** (ESI⁺): exact mass calculated for [M+H]⁺ (C₁₉H₁₈BrO₄) requires m/z 389.0388, found: m/z 389.0387.

4.11. 5-((2-methoxyphenyl)(phenyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2i**) [28]

White solid. **Yield:** 61.2 mg, 60%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). **¹H NMR** (400 MHz, CDCl₃, δ ppm): δ 7.40–7.26 (m, 5H), 7.23–7.17 (m, 1H), 6.89–6.81 (m, 3H), 5.45 (d, $J = 2.8$ Hz, 1H), 4.28 (d, $J = 2.8$ Hz, 1H), 3.83 (s, 3H), 1.72 (s, 3H), 1.70 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃, δ ppm): 165.5, 164.7, 156.0, 140.7, 130.7, 129.4, 128.9, 128.6, 127.9, 127.1, 120.5, 109.8, 104.5, 54.8, 50.4, 44.8, 28.2, 27.5.

4.12. 2,2-Dimethyl-5-(phenyl(*o*-tolyl)methyl)-1,3-dioxane-4,6-dione (**2j**)

White solid (mp: 121 °C). **Yield:** 79.7 mg, 82%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). **¹H NMR** (400 MHz, CDCl₃) δ 7.53–7.49 (m, 1H), 7.27–7.13 (m, 8H), 5.56 (d, $J = 2.8$ Hz, 1H), 4.26 (d, $J = 2.8$ Hz, 1H), 2.19 (s, 3H), 1.77 (s, 3H), 1.64 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 165.6, 164.1, 139.3, 138.6, 136.8, 130.8, 129.0, 128.2, 128.19, 127.3, 126.9, 126.3, 105.0, 51.3, 45.5, 28.3, 27.5, 20.0. **IR** ($\nu_{\max}$, cm⁻¹): 2919, 1787, 1746, 1316, 1293, 1194, 1175, 1055, 1010, 897, 743. **HRMS** (ESI⁺): exact mass calculated for [M+H]⁺ (C₂₀H₂₁O₄) requires m/z 325.1440, found: m/z 325.1449.

4.13. 2,2-Dimethyl-5-(phenyl(2-(trifluoromethyl)phenyl)methyl)-1,3-dioxane-4,6-dione (**2k**)

White solid (mp: 128 °C). **Yield:** 102.1 mg, 90%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). **¹H NMR** (400 MHz, CDCl₃) δ 7.81 (d, $J = 8.0$ Hz, 1H), 7.70 (d, $J = 7.6$ Hz, 1H), 7.52 (t, $J = 7.6$ Hz, 2H), 7.39 (t, $J = 7.6$ Hz, 1H), 7.31–7.21 (m, 2H), 7.07 (d, $J = 7.6$ Hz, 2H), 5.80 (d, $J = 3.5$ Hz, 1H), 4.18 (d, $J = 3.5$ Hz, 1H), 1.76 (s, 6H). **¹³C NMR** (100 MHz, CDCl₃, δ ppm): 164.3, 163.7, 139.6 (q, $J = 1.2$ Hz), 138.7, 135.6, 133.5, 131.8 (q, $J = 1.0$ Hz), 128.4, 128.2, 127.4, 127.0, 126.0 (q, $J = 6.1$), 124.6 (q, $J = 272.3$ Hz), 105.1, 51.0 (q, $J = 0.7$ Hz), 43.8 (q, $J = 1.9$ Hz), 28.4, 27.0. **IR** ($\nu_{\max}$, cm⁻¹): 2926, 1716, 1309, 1154, 1116, 1062, 1036, 768, 697, 598. **HRMS** (ESI⁺): exact mass calculated for [M+H]⁺ (C₂₀H₁₈F₃O₄) requires m/z 379.1157, found: m/z 379.1155.

4.14. 2,2-Dimethyl-5-(phenyl(2,4,6-trimethoxyphenyl)methyl)-1,3-dioxane-4,6-dione (**2l**)

Orange solid (mp: 121 °C). **Yield:** 34.8 mg, 29%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 70:30). **¹H NMR** (400 MHz, CDCl₃) δ 7.36–7.32 (m, 2H), 7.25–7.20 (m, 2H), 7.17–7.11 (m, 1H), 6.10 (s, 2H), 5.31 (d, $J = 8.9$ Hz, 1H), 4.82 (d, $J = 8.9$ Hz, 1H), 3.76 (s, 3H), 3.73 (s, 6H), 1.81 (s, 3H), 1.77 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 164.7, 163.6, 160.0, 158.3, 142.1, 128.0, 127.7, 126.1, 109.3, 104.4, 91.3, 55.5, 55.2, 49.7, 39.8, 28.7, 27.2. **IR** ($\nu_{\max}$, cm⁻¹): 2941, 2836, 1745, 1593, 1291, 1203, 1126, 827, 697, 542. **HRMS** (ESI⁺): exact mass calculated for [M+H]⁺ (C₂₂H₂₅O₇) requires m/z 401.1600, found: m/z 401.1598.

4.15. 2,2-Dimethyl-5-(naphthalen-1-yl(phenyl)methyl)-1,3-dioxane-4,6-dione (**2m**)

Light brown solid (mp: 132 °C). **Yield:** 45.4 mg, 42%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). **¹H NMR** (400 MHz, CDCl₃, δ ppm): δ 7.98–7.86 (m, 2H), 7.82 (d, $J = 8.4$ Hz, 1H), 7.63 (d, $J = 7.2$ Hz, 1H), 7.51–7.42 (m, 3H), 7.25–7.18 (m, 5H), 6.20 (d, $J = 2.6$ Hz, 1H), 4.40 (d, $J = 2.6$ Hz, 1H), 1.76 (s, 3H), 1.65 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃, δ ppm): 165.7, 164.0, 139.5, 136.0, 133.9, 131.3, 129.1 (2C), 128.4, 128.2, 127.13, 127.11, 126.6, 125.6, 125.2, 123.2, 105.1, 51.1, 45.2, 28.2, 27.6. **IR** ($\nu_{\max}$, cm⁻¹): 2923, 1705, 1494, 1306, 1198, 926, 779, 731, 601. **HRMS** (ESI⁺): exact mass calculated for [M+Na]⁺ (C₂₃H₂₀O₄Na) requires m/z 383.1259, found: m/z 383.1243.

4.16. 5-(Furan-2-yl(phenyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2n**)

Light brown solid (mp: 119 °C). **Yield:** 63.0 mg, 70%. Purified by

flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). **¹H NMR** (400 MHz, CDCl₃) δ 7.44–7.40 (m, 2H), 7.35–7.26 (m, 3H), 7.22 (dd, *J* = 5.1, 1.1 Hz, 1H), 7.01 (dt, *J* = 3.5, 1.1 Hz, 1H), 6.97–6.94 (m, 1H), 5.59 (d, *J* = 2.5 Hz, 1H), 4.24 (d, *J* = 2.5 Hz, 1H), 1.74 (s, 3H), 1.54 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 164.2, 164.1, 142.7, 139.4, 129.0, 128.5, 127.6, 127.3, 126.7, 125.0, 105.2, 52.5, 44.8, 28.2, 27.5. **IR** ($\nu_{\max}$, cm⁻¹): 3002, 1781, 1743, 1602, 1441, 1383, 1205, 1078, 1011, 896. **HRMS** (ESI⁺): exact mass calculated for [M+Na]⁺ (C₁₇H₁₆O₅Na) requires *m/z* 323.0895, found: *m/z* 323.0892.

4.17. 2,2-Dimethyl-5-(phenyl(thiophen-2-yl)methyl)-1,3-dioxane-4,6-dione (**2o**)

Dark orange solid (mp: 111 °C). **Yield**: 82.5 mg, 87%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). **¹H NMR** (400 MHz, CDCl₃) δ 7.49–7.45 (m, 2H), 7.37–7.35 (m, 1H), 7.35–7.33 (m, 1H), 7.33–7.31 (m, 2H), 6.34–6.32 (m, 1H), 6.06–6.05 (m, 1H), 5.28 (d, *J* = 3.0 Hz, 1H), 4.26 (d, *J* = 3.0 Hz, 1H), 1.73 (s, 3H), 1.51 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 164.2, 164.1, 153.1, 141.4, 137.3, 129.7, 128.6, 128.0, 110.6, 108.0, 105.1, 50.4, 44.0, 28.2, 27.6. **IR** ($\nu_{\max}$, cm⁻¹): 2923, 2841, 1751, 1415, 1215, 1007, 696. **HRMS** (ESI⁺): exact mass calculated for [M+H]⁺ (C₁₇H₁₇O₄S) requires *m/z* 317.0848, found: *m/z* 317.0845.

4.18. 2,2-Dimethyl-5-(3-methyl-1-phenylbutyl)-1,3-dioxane-4,6-dione (**2p**)

White solid (mp: 121 °C). **Yield**: 68.8 mg, 79%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 90:10). **¹H NMR** (400 MHz, CDCl₃) δ 7.36–7.18 (m, 5H), 3.87 (ddd, *J* = 10.1, 5.8, 3.1 Hz, 1H), 3.64 (d, *J* = 3.1 Hz, 1H), 2.30 (ddd, *J* = 13.9, 10.2, 5.5 Hz, 1H), 1.81–1.72 (m, 2H), 1.62 (s, 3H), 1.11 (s, 3H), 0.92 (dd, *J* = 6.6, 2.5 Hz, 6H). **¹³C NMR** (100 MHz, CDCl₃) δ 165.8, 164.7, 139.4, 129.1, 128.7, 127.6, 105.4, 51.5, 44.1, 41.2, 28.4, 28.0, 25.6, 23.1, 21.7. **IR** ($\nu_{\max}$, cm⁻¹): 2971, 1772, 1278, 1204, 1085, 1006, 726, 707, 545. **HRMS** (ESI⁺): exact mass calculated for [M+H]⁺ (C₁₇H₂₃O₄) requires *m/z* 291.1596, found: *m/z* 291.1591.

4.19. 2,2-Dimethyl-5-(2-methyl-1-phenylpropyl)-1,3-dioxane-4,6-dione (**2q**)

White solid (mp: 117 °C). **Yield**: 61.3 mg, 74%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 90:10). **¹H NMR** (400 MHz, CDCl₃) δ 7.38–7.12 (m, 5H), 3.89 (d, *J* = 3.2 Hz, 1H), 3.29 (dd, *J* = 11.4, 3.2 Hz, 1H), 2.86–2.70 (m, 1H), 1.60 (s, 3H), 1.21 (d, *J* = 6.5 Hz, 3H), 0.99 (s, 3H), 0.76 (d, *J* = 6.6 Hz, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 166.5, 164.8, 139.8, 129.1, 128.7, 127.6, 105.5, 54.6, 48.7, 29.1, 28.6, 27.9, 21.6, 21.2. **IR** ($\nu_{\max}$, cm⁻¹): 2966, 1771, 1732, 1287, 1204, 1085, 1002, 726, 705, 541. **HRMS** (ESI⁺): exact mass calculated for [M+H]⁺ (C₁₆H₂₁O₄) requires *m/z* 277.1440, found: *m/z* 277.1434.

4.20. 5-((4-chlorophenyl)(4-methoxyphenyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2r**) [30]

Light brown solid. **Yield**: 76.5 mg, 68%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). **¹H NMR** (400 MHz, CDCl₃) δ 7.28–7.18 (m, 6H), 6.88–6.81 (m, 2H), 5.31 (d, *J* = 2.6 Hz, 1H), 4.25 (d, *J* = 2.6 Hz, 1H), 3.79 (s, 3H), 1.75 (s, 3H), 1.56 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 164.59, 164.56, 158.7, 138.9, 132.8, 131.4, 130.4, 130.38, 128.4, 113.9, 105.2, 55.2, 51.1, 47.7, 28.2, 27.5.

4.21. 5-((3-methoxyphenyl)(4-methoxyphenyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2s**)

Yellow solid (mp: 146 °C). **Yield**: 74.4 mg, 67%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 75:25). **¹H NMR** (400 MHz, CDCl₃) δ 7.26–7.20 (m, 3H), 6.91–6.77 (m, 5H), 5.31 (d, *J* = 2.6 Hz, 1H), 4.26 (d, *J* = 2.6 Hz, 1H), 3.79 (s, 3H), 3.76 (s, 3H), 1.75 (s, 3H), 1.54 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 164.9, 164.6, 159.5, 158.5, 142.0, 131.8, 130.4, 129.3, 121.2, 115.1, 113.7, 112.1, 105.0, 55.14, 55.11, 51.2, 48.4, 28.2, 27.5. **IR** ($\nu_{\max}$, cm⁻¹): 2929, 1735, 1609, 1585, 1512, 1489, 1454, 1180, 791. **HRMS** (ESI⁺): exact mass calculated for [M+Na]⁺ (C₂₁H₂₂O₆Na) requires *m/z* 393.1314, found: *m/z* 393.1305.

4.22. 5-((4-methoxyphenyl)(*o*-tolyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2t**)

Light brown solid (mp: 120 °C). **Yield**: 69.0 mg, 65%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). **¹H NMR** (400 MHz, CDCl₃) δ 7.54–7.50 (m, 1H), 7.25–7.13 (m, 3H), 7.11–7.05 (m, 2H), 6.80–6.74 (m, 2H), 5.50 (d, *J* = 2.7 Hz, 1H), 4.24 (d, *J* = 2.7 Hz, 1H), 3.75 (s, 3H), 2.17 (s, 3H), 1.76 (s, 3H), 1.63 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 165.7, 164.1, 158.4, 139.0, 136.7, 131.1, 130.8, 130.3, 127.9, 127.1, 126.2, 113.5, 104.9, 55.1, 51.3, 45.0, 28.2, 27.5, 20.0. **IR** ($\nu_{\max}$, cm⁻¹): 2931, 1743, 1610, 1511, 1461, 1383, 1178, 1033, 1011. **HRMS** (ESI⁺): exact mass calculated for [M+H]⁺ (C₂₁H₂₂O₅Na) requires *m/z* 377.1365, found: *m/z* 377.1363.

4.23. 5-(bis(4-methoxyphenyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2u**) [31]

White solid. **Yield**: 85.6 mg, 77%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 75:25). **¹H NMR** (400 MHz, CDCl₃) δ 7.25–7.19 (m, 4H), 6.85–6.80 (m, 4H), 5.28 (d, *J* = 2.7 Hz, 1H), 4.25 (d, *J* = 2.7 Hz, 1H), 3.77 (s, 6H), 1.72 (s, 3H), 1.51 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 164.8, 158.4, 132.3, 130.2, 113.6, 105.0, 55.1, 51.3, 47.9, 28.2, 27.6.

4.24. 5-((2-methoxyphenyl)(4-methoxyphenyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2v**)

Brown solid (mp: 149 °C). **Yield**: 78.8 mg, 71%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 75:25). **¹H NMR** (400 MHz, CDCl₃) δ 7.34–7.29 (m, 2H), 7.22–7.16 (m, 1H), 6.94–6.81 (m, 5H), 5.38 (d, *J* = 2.6 Hz, 1H), 4.28 (d, *J* = 2.6 Hz, 1H), 3.82 (s, 3H), 3.79 (s, 3H), 1.71 (s, 3H), 1.69 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 165.5, 164.8, 158.5, 156.0, 132.6, 130.6, 130.5, 129.1, 127.8, 120.5, 113.9, 109.8, 104.4, 55.1, 54.8, 50.6, 44.1, 28.1, 27.5. **IR** ($\nu_{\max}$, cm⁻¹): 2870, 1735, 1512, 1492, 1181, 1119, 1028, 754. **HRMS** (ESI⁺): exact mass calculated for [M+Na]⁺ (C₂₁H₂₂O₆Na) requires *m/z* 393.1314, found: *m/z* 393.1306.

4.25. 5-((4-methoxyphenyl)(*p*-tolyl)methyl)-2,2-dimethyl-1,3-dioxane-4,6-dione (**2w**)

Yellow solid (mp: 170 °C). **Yield**: 86.12 mg, 81%. Purified by flash chromatography using hexane:AcOEt (gradient from hexane/AcOEt = 90:10 to hexane/AcOEt = 80:20). **¹H NMR** (400 MHz, CDCl₃) δ 7.25–7.21 (m, 2H), 7.21–7.16 (m, 2H), 7.13–7.08 (m, 2H), 6.85–6.80 (m, 2H), 5.29 (d, *J* = 2.7 Hz, 1H), 4.26 (d, *J* = 2.7 Hz, 1H), 3.78 (s, 3H), 2.31 (s, 3H), 1.73 (s, 3H), 1.52 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 164.9, 164.7, 158.5, 137.3, 136.6, 132.1, 130.4, 129.0, 128.8, 113.7, 105.0, 55.2, 51.3, 48.2, 28.2, 27.6, 21.0. **IR** ($\nu_{\max}$, cm⁻¹): 2994,

1742, 1610, 1541, 1295, 1011. **HRMS** (ESI⁺): exact mass calculated for [M+H]⁺ (C₂₁H₂₂O₅Na) requires *m/z* 377.1365, found: *m/z* 377.1354.

4.26. Procedure for the decarboxylation reaction

Following the method developed by Frost [32], in a 25 mL flask coupled with reflux condenser were added 0.125 mL of concentrated HCl, 5 mL of a 10:1 mixture of DMF/H₂O and the arylated product 2 (0.3 mmol). The reaction mixture was heated to 100 °C and stirred for 1 h. After that, the system was cooled to room temperature and the organic phase was extracted with ethyl acetate (3 × 10 mL). The organic phase was dried with Na₂SO₄ and the product was concentrated under reduced pressure. The product is obtained pure and does not require further purification steps.

4.27. 3-(4-chlorophenyl)-3-phenylpropanoic acid (**3a**) [33]

White solid. **Yield**: 74.3 mg, 95%. **¹H NMR** (400 MHz, CDCl₃) δ 7.39–7.08 (m, 9H), 4.49 (t, *J* = 7.9 Hz, 1H), 3.05 (d, *J* = 7.9 Hz, 2H). **¹³C NMR** (100 MHz, CDCl₃) δ 177.0, 142.7, 141.7, 132.4, 129.0, 128.7, 128.73, 127.5, 126.8, 46.0, 40.2.

4.28. 3-(4-methoxyphenyl)-3-phenylpropanoic acid (**3b**) [32]

White solid. **Yield**: 73.8 mg, 96%. **¹H NMR** (400 MHz, CDCl₃) δ 7.24–7.01 (m, 7H), 6.78–6.68 (m, 2H), 4.39 (t, *J* = 7.9 Hz, 1H), 3.67 (s, 3H), 2.96 (d, *J* = 7.9 Hz, 2H). **¹³C NMR** (100 MHz, CDCl₃) δ 177.6, 158.1, 143.6, 135.4, 128.55, 128.54, 127.5, 126.5, 114.0, 55.2, 45.8, 40.6.

4.29. Procedure for the synthesis of amide **3c**

A solution of **2c** (64.9 mg, 0.2 mmol) and benzylamine (22 μL, 0.2 mmol) in CH₃CN (2 mL) was stirred at 90 °C for 16 h. The solvent was removed under reduced pressure. The remaining crude was purified by column chromatography (in DCM) on silica flash to afford the *N*-benzyl-3-phenyl-3-(*p*-tolyl)propanamide **3c** (53.4 mg, 81%) as a yellow solid.

4.30. *N*-benzyl-3-phenyl-3-(*p*-tolyl)propanamide (**3c**) [34]

White solid. **Yield**: 73.8 mg, 96%. **¹H NMR** (400 MHz, CDCl₃) δ 7.34–7.05 (m, 12H), 6.92–6.85 (m, 2H), 5.53 (s, 1H), 4.56 (t, *J* = 7.9 Hz, 1H), 4.30 (d, *J* = 5.6 Hz, 2H), 2.92 (d, *J* = 7.9 Hz, 2H), 2.31 (s, 3H). **¹³C NMR** (100 MHz, CDCl₃) δ 171.0, 143.8, 140.5, 137.9, 136.0, 129.3, 128.6, 128.4, 127.7, 127.6, 127.4, 127.2, 126.4, 47.1, 43.5, 43.4, 21.0.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tet.2020.130967>.

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